

Multiple Choice Questions (MCQs)

1. What does AOP stand for?

- A) Application-Oriented Programming
- B) Aspect-Oriented Programming
- C) Aspect-Optimized Programming
- D) Application-Optimized Programming

Answer: B) Aspect-Oriented Programming

2. Which of the following is a cross-cutting concern?

- A) Business logic
- B) Data access
- C) Logging
- D) User interface

Answer: C) Logging

3. What annotation is used to define an aspect in Spring AOP?

- A) @Component
- B) @Service
- C) @Aspect
- D) @Controller

Answer: C) @Aspect

4. What is a join point in Spring AOP?

- A) A point where two aspects meet
- B) A point during the execution of a program, such as method execution
- C) A point where advice is applied
- D) A point where the application is compiled

Answer: B) A point during the execution of a program, such as method execution

5. Which type of advice runs before the join point?

- A) After Advice
- B) Around Advice
- C) Before Advice
- D) After Returning Advice

Answer: C) Before Advice

6. What is the purpose of a pointcut in Spring AOP?

- A) To define where advice should be applied
- B) To execute the advice
- C) To create a proxy object
- D) To handle exceptions

Answer: A) To define where advice should be applied

7. Which annotation is used to define "after" advice?

- A) @Before
- B) @After
- C) @Around

- D) @AfterReturning

Answer: B) @After

8. What does the @AfterReturning annotation do?

- A) It runs after a method throws an exception
- B) It runs after a method returns successfully
- C) It runs before a method execution
- D) It runs around a method execution

Answer: B) It runs after a method returns successfully

9. What is the target object in Spring AOP?

- A) The object that contains the aspect
- B) The object being advised by one or more aspects
- C) The object that executes the advice
- D) The object that defines the pointcut

Answer: B) The object being advised by one or more aspects

10. Which of the following is NOT a type of advice in Spring AOP?

- A) Before Advice
- B) After Advice
- C) During Advice
- D) Around Advice

Answer: C) During Advice

11. What is weaving in the context of Spring AOP?

- A) The process of compiling the application
- B) The process of linking aspects with application types to create an advised object
- C) The process of executing the advice
- D) The process of defining pointcuts

Answer: B) The process of linking aspects with application types to create an advised object

12. Which type of proxy does Spring use when the target object implements one or more interfaces?

- A) CGLIB Proxy
- B) JDK Dynamic Proxy
- C) Static Proxy
- D) Custom Proxy

Answer: B) JDK Dynamic Proxy

13. What is the purpose of the @Pointcut annotation?

- A) To define a join point
- B) To define a pointcut expression
- C) To execute advice
- D) To create a proxy

Answer: B) To define a pointcut expression

14. Which of the following is an example of around advice?

- A) Logging before a method execution
- B) Logging after a method execution

- C) Logging before and after a method execution
- D) Logging only if a method throws an exception

Answer: C) Logging before and after a method execution

15. In which phase does Spring AOP perform weaving?

- A) Compile time
- B) Load time
- C) Runtime
- D) All of the above

Answer: C) Runtime

16. What is the main benefit of using AOP in an application?

- A) Increased performance
- B) Separation of cross-cutting concerns from business logic
- C) Simplified code structure
- D) Enhanced security

Answer: B) Separation of cross-cutting concerns from business logic

17. Which of the following annotations is used to define "around" advice?

- A) @Before
- B) @After
- C) @Around
- D) @AfterThrowing

Answer: C) @Around

18. What is the role of the @Component annotation in an aspect class?

- A) It defines the aspect
- B) It marks the class as a Spring-managed bean
- C) It indicates the class is a service
- D) It defines a pointcut

Answer: B) It marks the class as a Spring-managed bean

19. Which of the following is a valid pointcut expression?

- A) `execution(* com.example.service.*.*(..))`
- B) `execution(com.example.service.*.*(..))`
- C) `execution(* com.example.service.*)`
- D) `execution(* com.example.service.*.*.void(..))`

Answer: A) `execution(* com.example.service.*.*(..))`

20. What happens if a method annotated with @AfterThrowing throws an exception?

- A) The exception is ignored
- B) The advice runs after the exception is thrown
- C) The application crashes
- D) The method is retried

Answer: B) The advice runs after the exception is thrown